# StepUp Project Report


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# Overview

StepUp is a UCS503 project developed by Team IRIS at Thapar Institute of Engineering and Technology. The project combines a student fitness application with two technical tracks: activity integrity detection and constraint-based sport group matching. Its purpose is to encourage regular physical activity while making recorded progress and group participation more trustworthy.

## 1. Project Background and Problem Statement

The project addresses two practical problems observed in campus fitness systems. First, imported step records may include duplicate synchronization, phone shaking, irregular motion or vehicle travel. Second, students often form sport groups through informal messages without dependable skill, schedule or attendance matching.

- **Activity data integrity:** Step counts should represent genuine walking or running before they affect rankings, rewards or challenge results.
- **Sport group formation:** Students need compatible groups based on sport, skill band, time overlap, group size and coarse location.
- **Academic context:** The system is designed as a university project for measurable experimentation, modular development and final demonstration.

## 2. Objectives and Scope

StepUp converts the problem statement into four concrete project objectives:

- **Integrity objective:** Build an on-device classifier and rule-based baseline that separate genuine activity from suspicious records.
- **Matching objective:** Develop a constraint-based sport group matching engine with reliability and preference scoring.
- **Application objective:** Deliver a Flutter experience for profiles, activity summaries, challenges, leaderboards, rewards, sessions and notifications.
- **Scope boundary:** The work focuses on integrity, matching and supporting interfaces; it is not intended to replace a general-purpose medical or fitness tracker.

## 3. Incremental Development Progress

Development follows a 15-week plan in which each phase produces a testable part of the system. The first thirteen weeks are recorded as complete in the project website, placing the project at 87 percent progress before final integration and deployment.

| Phase | Planned Focus | Current Outcome |
|---|---|---|
| **Weeks 1-3** | Ideation, requirements and architecture | Problem tracks defined; architecture and technology stack selected. |
| **Weeks 4-6** | Application foundation | Navigation, authentication, student profiles and activity dashboard completed. |
| **Weeks 7-9** | Step integrity and Health Connect | Sensor input, six-layer validation and Health Connect synchronization completed. |
| **Weeks 10-13** | Social fitness and rewards | Leaderboard, challenges, achievements, wallet, sessions and notifications completed. |
| **Weeks 14-15** | Integration, testing and deployment | Scheduled for final integration, optimization, documentation and demonstration. |

## 4. Current Status and Next Steps

- **Current position:** Week 13 is complete, including the wallet, activity session history and notifications for goals, challenges and achievements.
- **Week 14:** Integrate the major modules, test end-to-end functionality and resolve defects found across the Flutter app, integrity pipeline and backend services.
- **Week 15:** Optimize performance, refine the interface, prepare technical documentation and finalize the project demonstration.
- **Evidence requirement:** Final claims will be supported by recorded test results rather than planned metrics alone.

# System Design and Development

StepUp uses a layered design that separates device data collection, activity validation, user-facing features and server-side coordination. This separation allows each technical track to be evaluated independently while still contributing to a single student fitness experience.

## 1. From Requirements to Architecture

The selected architecture maps the project requirements to a clear flow of data and responsibilities:

- **Device sources:** Android Health Connect provides available step records and origin information, while the accelerometer, gyroscope and hardware step counter provide motion evidence.
- **On-device processing:** A six-layer integrity pipeline checks source trust, duplicates, shake noise, step peaks, walking periodicity, vehicle patterns and cross-validation limits.
- **Application layer:** Flutter presents the dashboard, profiles, challenges, leaderboard, wallet, rewards and activity sessions.
- **Backend and data:** REST services support authentication, challenge coordination, rankings and rewards, with PostgreSQL planned for persistent project data.

## 2. Activity Integrity Detection

The integrity track is intended to prevent unverified activity from directly influencing social and reward features.

- **Target classes:** Genuine walking, genuine running, phone shaking or fake steps, duplicate synchronization and vehicle travel.
- **Decision method:** The project begins with interpretable rules and then evaluates a compact on-device machine-learning model using team-device data.
- **Leakage control:** Training and testing are split by participant rather than by session so the evaluation does not reward memorization of one person's motion pattern.

## 3. Sport Group Matching

The matching track treats group formation as a constrained optimization problem instead of a simple contact list.

- **Hard constraints:** Same sport, compatible skill band, overlapping availability, acceptable group size and coarse location compatibility.
- **Optimization factors:** Skill fit, attendance reliability and stated preferences are combined after the hard constraints are satisfied.

## 4. Privacy Data Handling and Constraints

Because StepUp processes activity and participation data, privacy and technical constraints are treated as design requirements rather than later additions.

- **Privacy controls:** Explicit consent, limited retention, on-device processing where possible, coarse location, no continuous background tracking, deletion controls and visibility settings.
- **Device variability:** Sensor behavior and Health Connect compatibility may differ across Android devices, so thresholds and test results must be reported by device context.
- **Dataset limitations:** A team-device dataset is suitable for a project prototype but may not represent broader student populations; conclusions must remain proportional to the sample.
- **Model constraints:** On-device inference requires a small model, low latency and acceptable battery cost without weakening the rule-based fallback.

## 5. Team Roles and Project Management

The work is divided by subsystem while integration and weekly planning remain shared responsibilities.

- **Disha Goyal:** Team coordination, Flutter mobile architecture, navigation, Health Connect integration, documentation and website management.
- **Kashish Singhal:** Activity-integrity dataset, rule-based baseline, on-device model and evaluation experiments.
- **Anvi Nagpal:** Backend REST APIs, matching formulation, matching engine, evaluation logging and mobile integration.
- **Raghav Sharma:** Frontend interface, backend integration, data presentation, visual implementation and cross-platform interface testing.

## 6. Completion Plan and Expected Deliverables

- **Integrated prototype:** A working StepUp application connecting verified activity, student challenges, rankings, rewards and sport group features.
- **Technical evidence:** Integrity metrics, matching metrics, latency and model-size observations, device notes and documented limitations.
- **Academic package:** Final source code, project report, presentation, demonstration flow and deployment-ready supporting website.

# Implementation Details

The implementation is organized into modules that can be developed, tested and integrated independently. This structure also makes the final report traceable from objectives to evidence.

## 1. Technical System Architecture

The final architecture connects the mobile client, device data sources, integrity logic, backend services and stored project data.

- **Technology stack:** Flutter for the mobile application; Android Health Connect and motion sensors for activity input; an on-device rule and ML layer for validation; REST APIs and PostgreSQL for service and data functions; and a static web dashboard for documentation and progress visibility.
- **Data workflow:** Step deltas and motion evidence are collected with permission, validated on the device, credited only after integrity checks, then synchronized with features that require shared state such as rankings, challenges and rewards.

## 2. Module by Module Implementation

- **Module A  Flutter application**
    - **Details:** Navigation, authentication, profiles, dashboard, challenges, leaderboard, wallet, rewards, sessions and notifications.
- **Module B  Activity integrity engine**
    - **Details:** Source and duplicate checks, shake filtering, motion peak detection, walking validation, vehicle rejection and cross-validation of credited steps.
- **Module C  Matching and backend services**
    - **Details:** Constraint filtering, preference and reliability scoring, authentication, challenge services, logging and database integration.
- **Module D  Web documentation**
    - **Details:** Project overview, architecture, 15-week progress, team responsibilities, repository activity and presentation archive.

## 3. Evaluation and Validation

- **Integrity metrics:** Precision, recall, F1 score, false-positive rate, inference latency, model size and battery cost, reported with the tested device conditions.
- **Matching metrics:** Acceptance rate, fairness, stability, session completion and no-show rate, compared with random and first-fit baselines where data permits.
- **Functional testing:** End-to-end checks for permissions, synchronization, challenge flows, leaderboard updates, rewards, notifications and failure handling.

# Conclusion

StepUp brings together genuine activity validation, social fitness features and structured sport group formation in one student-focused system. The project has progressed through thirteen of fifteen planned weeks, with the major application, integrity and social modules recorded as complete. The remaining work is centered on integration, measurement, optimization and evidence-backed documentation.

The final project position can be summarized as follows:

1. **Implemented foundation:** Flutter screens, profiles, activity summaries, sensor integration, integrity layers, leaderboard, challenges, rewards, wallet, sessions and notifications.
2. **Immediate priority:** Connect the modules into a stable end-to-end build and resolve issues found during Week 14 testing.
3. **Required proof:** Record integrity, matching and functional test outcomes with device conditions and limitations before making final performance claims.
4. **Future extension:** A larger pilot, expanded activity classes, university scheduling integration, federated learning and formal validation of the reliability model.